GAME DESIGN

Road Safety Simulator

v4.0

DIVIJA KALRA

24CD3013

OKESH DAYMA

24CD3031

ZOYA ALTAF

24CD3052

Project Overview

What is it?

- A Python/Pygame driving simulation game
- Infinite procedurally generated road — no finish line
- Game ends only on collision — survive as long as possible
- Manual 6-speed gearbox with realistic physics
- Dynamic weather: Clear, Rain & Night modes
- AI driving coach powered by Claude API
- High score system saved to local file
- 8 selectable car models with unique traits

Tech Stack

Python 3	Core language
Pygame	Game engine & rendering
NumPy	Audio synthesis
Requests	Claude API calls

Goal

Build an engaging, educational driving simulator that teaches road safety rules — traffic lights, pedestrian crossings, speed limits, fuel management — through hands-on gameplay with realistic car physics.

Team Roles & Contributions

Okesh Dayma

24CD3031

Lead Developer

- Primary code author — wrote majority of the codebase
- Implemented all core game physics (gear system, tyre grip, weight transfer, momentum)
- Built the procedural road & chunk generation system
- Developed all 8 car models with unique draw routines
- Engineered the engine audio synthesis using NumPy
- Created the HUD, scoring, and high score system
- Integrated the Anthropic API for AI coach feature

Divija Kalra

24CD3013

Research Lead

- Researched Pygame library capabilities & audio APIs
- Studied real-world road safety rules and traffic laws for accuracy
- Researched physics models: gear ratios, torque curves, braking distance
- Investigated Anthropic Claude API integration methods
- Gathered reference data for car speed profiles & gear speeds
- Documented research findings to guide development decisions

Zoya Altaf

24CD3052

Co-Developer & Tester

- Assisted in game logic development and debugging
- Designed the car selection screen UI and flow
- Contributed to NPC car behaviour & traffic AI
- Tested gameplay across all 8 vehicles and weather modes
- Compiled project documentation and README
- Prepared the final project presentation (this PPT)

Key Game Features

Realistic Physics

Weight transfer on braking, tyre grip model, gear inertia, torque curves per gear, and speed-sensitive steering.

Infinite Road

Procedurally generated road with chunks, randomised traffic lights, zebra crossings, NPC cars, and scenery.

Dynamic Weather

Three weather modes — Clear, Rain (reduced grip, wet road), and Night (auto headlights, dark overlay).

8 Car Models

Sports, Sedan, SUV, Muscle Car, Hatchback, Pickup Truck, Supercar, and Van — each with unique physics.

AI Coach

Scrolling tip bar powered by Claude API. Gives context-aware driving tips based on the current situation.

High Score System

Score based on distance survived and time. Graded S/A/B/C. Persisted to local file between sessions.

Controls & Gameplay

Driving Controls

→ / D	Accelerate
← / A	Reverse
SPACE	Brake
W / ↑	Steer up (top lane)
S / ↓	Steer down (bottom lane)
Q	Shift gear UP
E	Shift gear DOWN
ESC	Quit game
R	Restart after crash

Gear Speed Caps

G1 ~40 km/h	**G2** ~68 km/h	**G3** ~94 km/h
G4 ~117 km/h	**G5** ~140 km/h	**G6** ~158 km/h

Traffic Rules & Penalties

• Run red light	Score penalty
• Go through amber	Score penalty
• Hit pedestrian	Score penalty
• Hit NPC car	GAME OVER
• Run out of fuel	GAME OVER

Code Architecture

PlayerCar

Core physics: throttle smoothing, gear torque curves, brake weight transfer, steering damping, fuel drain, skid mark generation, engine audio index.

NPC Cars

AI-driven cars with lane awareness. Stop at red lights, follow-distance braking, variable speeds. Randomly placed across both lanes.

GameWorld

Master class: manages camera, chunk spawning, collision detection, scoring, HUD rendering, penalty messages and the game over trigger.

Weather

Cycles through Clear → Rain → Night automatically. Affects road colour, grip, visibility, rain drop physics, and headlight activation.

TrafficLight

Full Green/Amber/Red cycle with countdown timer. Draws stop line, triggers pedestrian spawning, penalises the player for violations.

AICoach

Background thread calls Claude API with situation context. Falls back to hardcoded tips if no API key set. Scrolling ticker at screen bottom.

Challenges & What We Learned

Challenges

- Getting gear physics to feel realistic without being frustrating
- Infinite road generation without visible seams or stuttering
- Making NPC AI brake and react naturally to traffic lights
- Synthesising engine audio with NumPy at varying RPMs
- Balancing game difficulty — too easy or too punishing kills fun
- Running Claude API calls on a background thread without blocking gameplay
- Collision detection accuracy with varied car sizes

What We Learned

- Pygame game loop architecture and frame-rate management
- Real-time audio synthesis using NumPy sine waves
- Procedural content generation for infinite levels
- Integrating a REST API into a live game loop safely
- Designing intuitive car physics that feel good to play
- How traffic rules and road safety translate into game mechanics
- Team collaboration — splitting code, research, and testing effectively

Thank You!

Drive safe. Always.

Subject: GAME DESIGN

Divija Kalra — 24CD3013
Okesh Dayma — 24CD3031
Zoya Altaf — 24CD3052